

VidExpLainAgent



DARTMOUTH

Technigala

AUTHOR

Syed Ali Haider

Towards Textual and Auditory Theorem
Explanations from Educational Video

1. Problem

The acquisition of knowledge in STEM is strongly linked to the development of visual intuition, which BLV people cannot access

1. **Unscalable:** Manual annotation cannot keep up with the rapid growth of online STEM videos
2. **Static:** A single, pre-recorded description is inflexible and cannot adapt to the diverse needs.
3. **Non-Interactive:** Learners cannot ask clarifying questions about visuals.

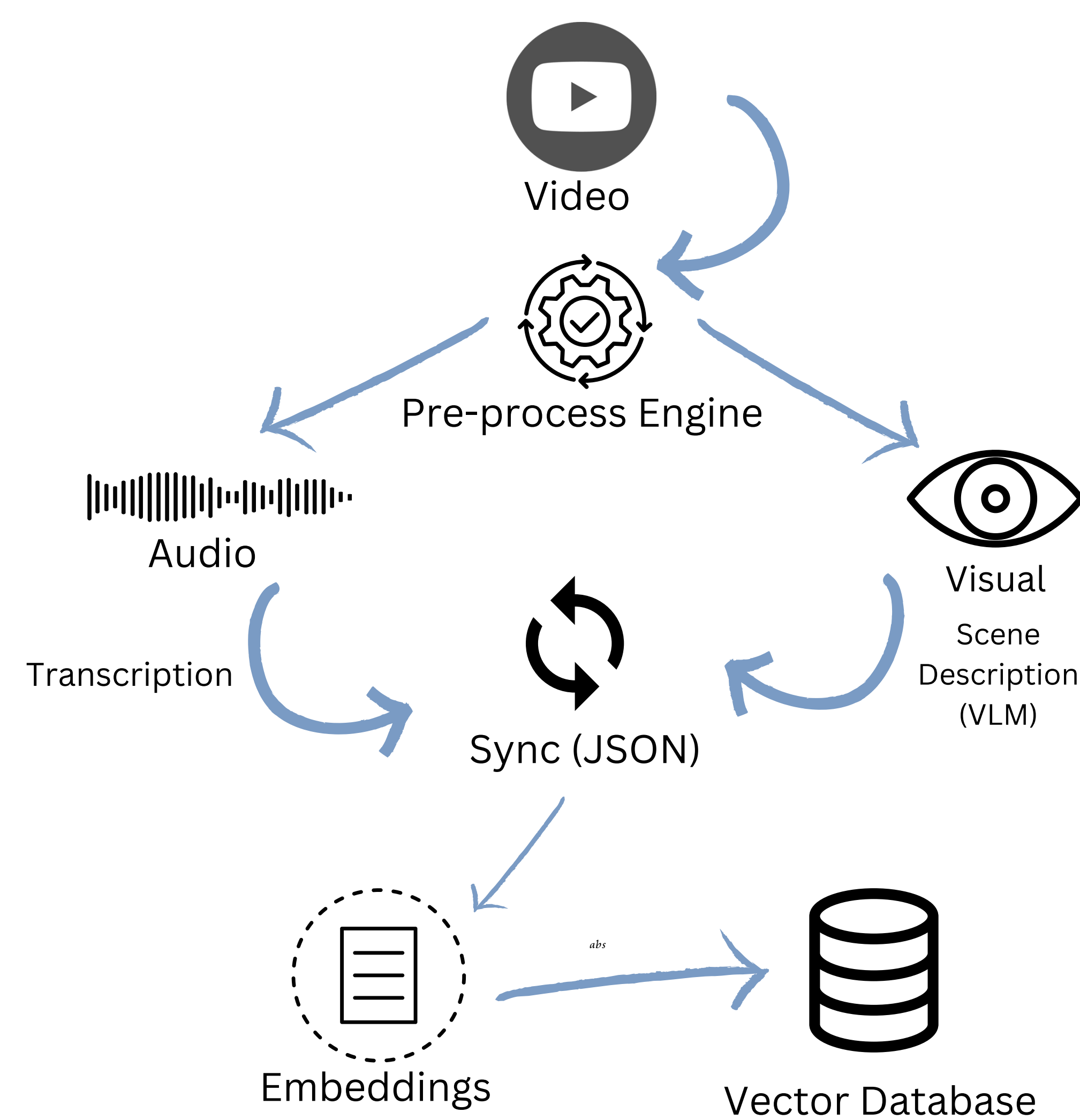
This creates a critical barrier for **BLV** users

2. What is Vid Explain Agent

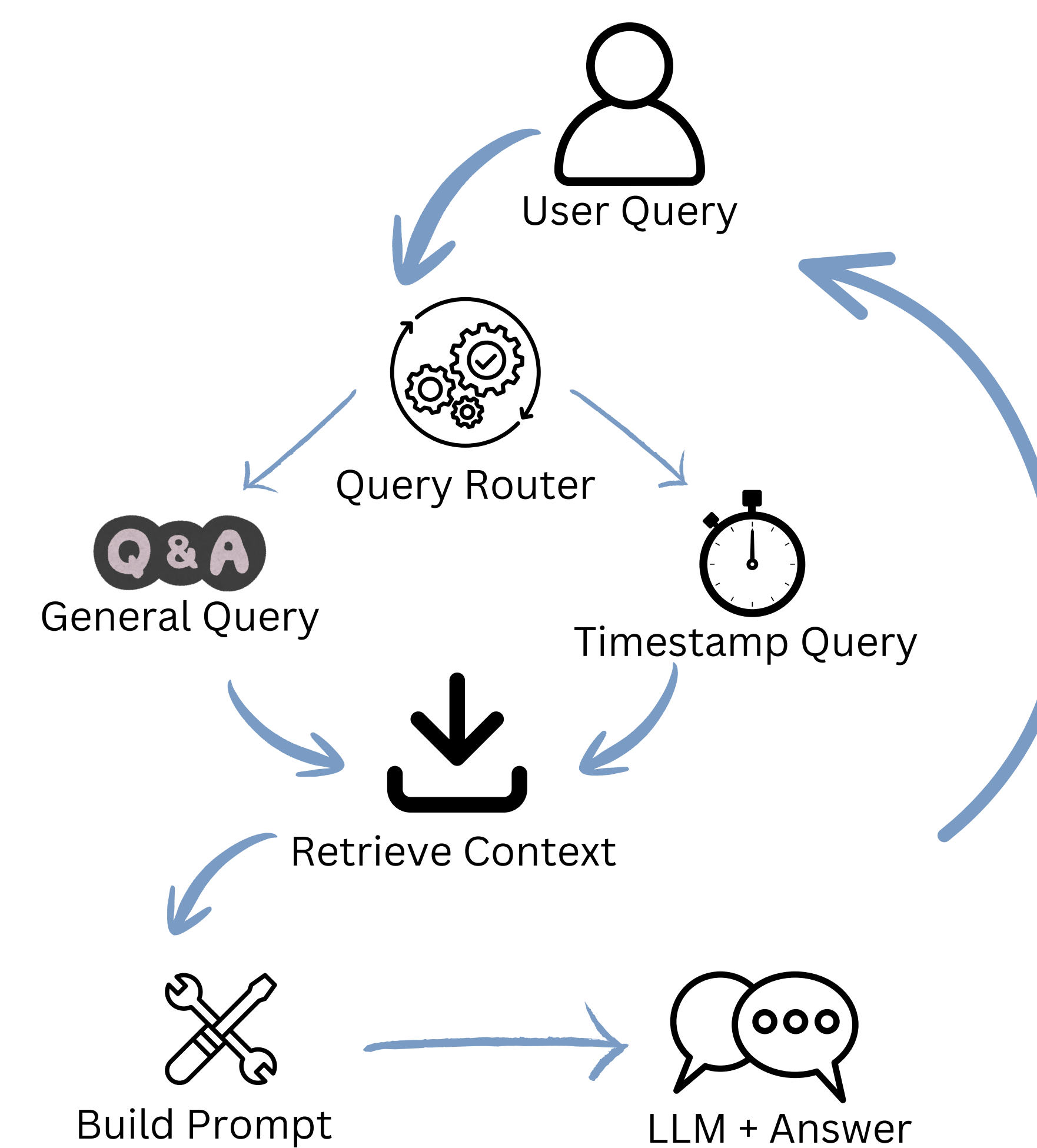
A multimodal Retrieval-Augmented Generation (RAG) system designed to provide interactive, on-demand access to educational video content for BLV learners.

3. Pipeline

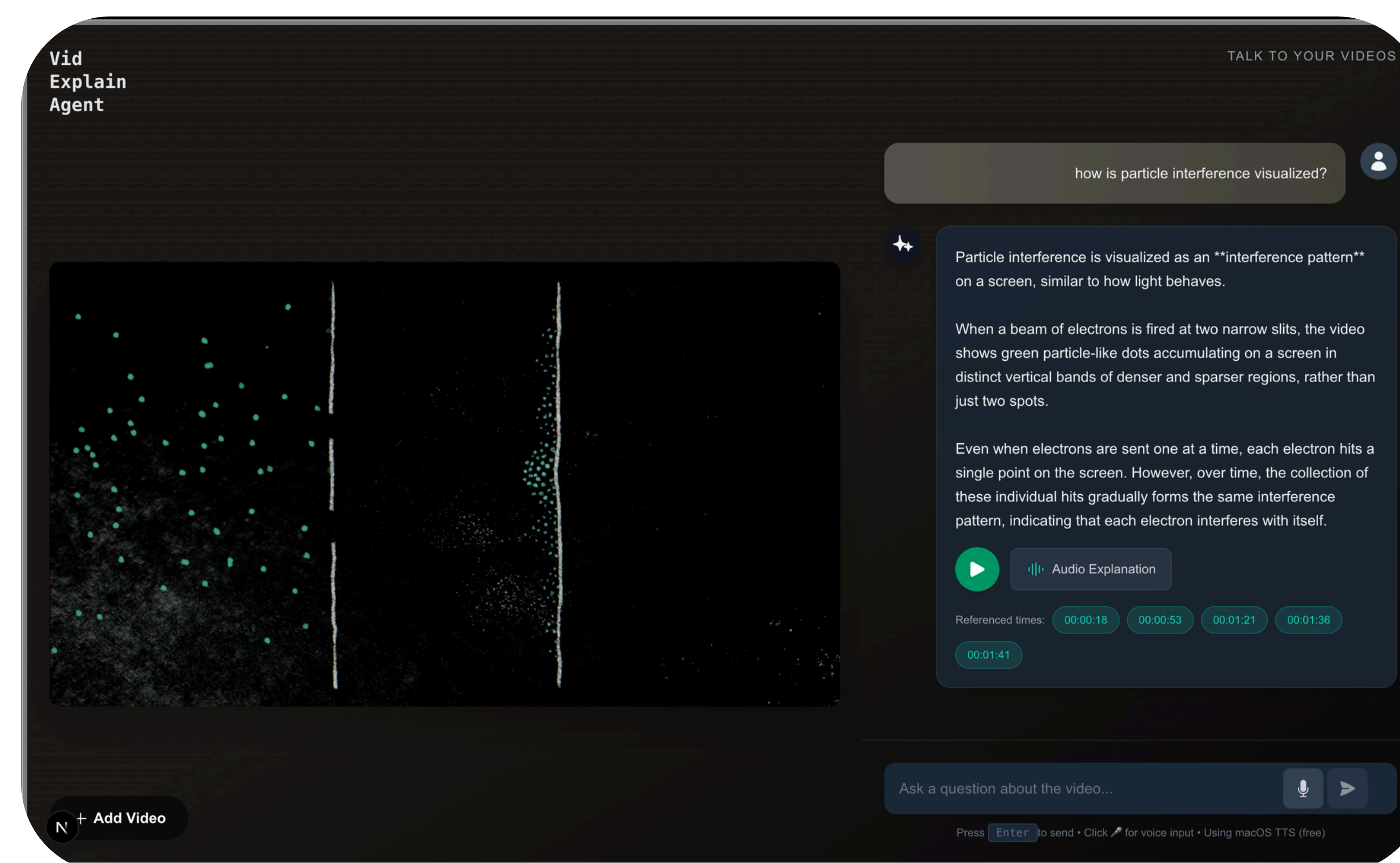
Phase 1: Ingestion & Indexing



Phase 2: Interactive Dialogue



3. Product Interface



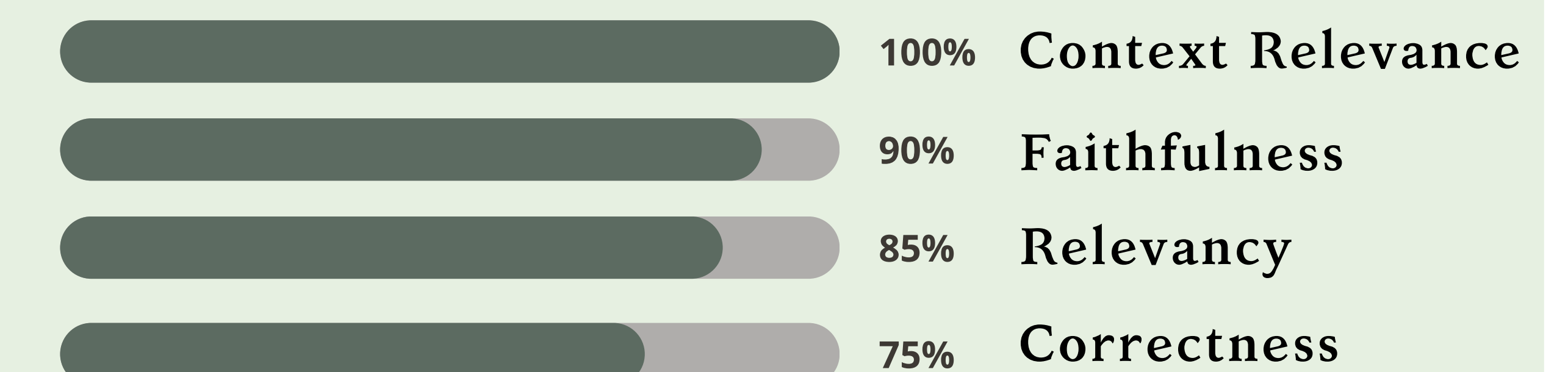
5. Results

0.588 *BERTScore F1*

Semantic Meaning > Lexical Match

The system achieves a strong BERTScore, indicating solid semantic understanding. The low BLEU score (0.045) is expected, as the VLM conveys accurate meaning using different vocabulary than human annotators, which is ideal for accessibility.

End-to-End (RAG) - 87.5% Pass Rate



The perfect 100% Context Relevance (retrieval) and 90% Faithfulness (no hallucination) are critical for a trustworthy educational tool.

6. Conclusion

VidExpLainAgent is a reliable and robust system, demonstrating that automated, high-quality accessibility for BLV users in STEM video is technically feasible.